

Management of Urinary Tract Infections

Questions	Please answer here	For pharmacist
Gender: What is the sex of the patient?	Female / Male or individuals who have undergone gender reassignment surgery	[Triggers: Male or individuals who have undergone gender reassignment surgery → Provide usual care and/or refer to GP]
Age: Is the patient aged between 18 and 65 years?	Age years	[Triggers: No (under 18 or over 65) → Provide usual care and/or refer to GP]
Consent: Does the patient consent? * (*The patient must be physically present in the pharmacy)	No / Yes	[Triggers: No → Provide usual care and/or refer to GP]
Symptoms: Is the patient presenting with 2 or more symptoms of acute cystitis? * *If vaginal discharge is absent and at least two symptoms of cystitis are present (i.e. acute dysuria, frequency, urgency), the probability of cystitis is greater than 90%	No / Yes ○ Dysuria ○ Urinary frequency ○ Urinary urgency ○ Suprapubic pain or discomfort	[Triggers: No (presenting with only 1 symptom of acute cystitis or atypical symptoms) → Provide usual care and/or refer to GP as patient may benefit from laboratory investigations]
Differential diagnosis: Do symptoms of medical history suggest a cause other than acute cystitis	Yes / No (e.g. vaginal discharge, or vulval and vaginal discomfort) → (e.g. vaginal thrush, bacterial vaginosis, vaginitis or vulvitis, chlamydia, gonorrhoea, trichomoniasis or other)	[Triggers: Consider appropriateness of S3 Pharmacist Only thrush treatment or refer to GP for investigation of other causes (e.g. infections such as bacterial vaginosis or herpes, or skin conditions)]
Red flag Symptoms: Does the patient have signs and symptoms indicating pyelonephritis ? (e.g. fever >38°C, chills, nausea, vomiting, flank pain)	No / Yes ○ Fever >38°C ○ Chills ○ Nausea ○ Vomiting ○ Flank pain	[Triggers: Yes → Immediate referral to Emergency Department]
Reg flag medical Hx: Does the patient have any of the following?	No / Yes ○ Pregnancy ○ Less than 6 weeks post-partum ○ Gross haematuria ○ Cerebral palsy or other neurological diseases that may affect bladder function ○ Recent insertion of IUD (within the last 3 months) ○ Urinary catheterisation (in situ or in the last 48 hours) ○ Presence of history of anatomical or functional abnormalities of the urinary tract including: (Urinary tract obstruction, Urinary tract abnormality, Urolithiasis, Nephrostomy tube, Ureteral stent, Spinal cord injury)	[Triggers: Yes → Immediate referral to GP Symptomatic treatment with nonprescription medicines (if appropriate) and self-care advice may be provided by pharmacist prior to review by GP]
Risks: Does the patient report any of the following?	No / Yes ○ Having diabetes or using medicines that increase the risk of UTI (e.g. SGLT2 inhibitors) ○ Reoccurrence of UTI symptoms within 2 weeks of completing appropriate antimicrobial treatment for a previous episode ○ Symptoms of UTI persisting 48-72 hours after starting appropriate antibiotic treatment ○ 2 or more symptomatic UTIs in previous 6 months OR 3 or more in previous 12 months ○ Long-term inpatient care (including residential aged care facilities)	[Triggers: Yes → Provide usual care and/or refer to GP as patient may benefit from laboratory investigations]
Soft triggers: Does the patient report / present with any of the following?	No / Yes ○ Risk factors for sexually transmitted infection (STI) including: ▪ Age < 30 years ▪ Previous STI ▪ Sexual contact without a condom/dental dam outside a mutually monogamous relationship ▪ A new sexual partner in last 60 days ▪ A sexual partner recently treated for an STI ▪ Sexual contact with a sex worker ○ History of pyelonephritis ○ Taking immunosuppressant medicines ○ Being immunocompromised ○ Asplenia ○ Renal disease or impairment ○ IUD in situ (greater than 3 months)	[Triggers: Yes → Antibiotic treatment by pharmacist may still be considered , concurrent to a referral to the GP, if clinically appropriate (e.g. possible delay in access to GP and no contraindications to antibiotics). → Provide usual care and/or refer to GP *** Refer patient for STI testing where risk factors are identified.

*** **Provide usual care** = including symptomatic treatment with non-prescription medicines (if appropriate) and self-care advice

PHARMACIST CARE

CONSERVATIVE MANAGEMENT	non-prescription medicines (as first line treatment for patients without immune compromise)	▪ Analgesia: ibuprofen 400mg orally, every 8 hours for up to 3 days. Maximum daily dose of 2.4g in 24 hours	If patient continues to have symptoms after 48 hours of conservative management, advise patient to return to pharmacy for review and consider antibiotic therapy .
	non-pharmacological and self-care advice	▪ Increasing water intake (up to 1.5 L daily) may reduce the risk of recurrent UTI in premenopausal females who have inadequate fluid intake (<1.5 L per day) ▪ CMI and/or a Self-Care Fact Card ▪ guidance on recognising signs and symptoms of worsening infection (e.g. fever 38°C or higher, rigors, loin or back pain, vomiting)	

The efficacy of **urinary alkalinising agents for the symptomatic treatment of acute cystitis has not been established in trials, but some patients find them useful for symptom relief.

****Cranberry products, ascorbic acid and methenamine hippurate** are not effective for the treatment of acute cystitis.

EMPIRICAL ANTIBIOTIC THERAPY

ANTIBIOTIC THERAPY Empirical antibiotic therapy, where indicated Dispense any medications (if supplied) via pharmacy dispensing software and label according to the legislative requirements	Nitrofurantoin (1st line) 100mg orally, every 6 hours for 5 days (Supply 20 capsules)	Contraindications: • Previous serious adverse reaction to nitrofurantoin • G6PD, enolase, or glutathione peroxidase deficiency (may lead to haemolytic anaemia) • Severe renal impairment • Avoid in breastfeeding if infant is under one month or has G6PD deficiency	Provide follow-up advice and expectations around duration of symptoms • Symptoms should respond to appropriate antibiotic treatment within 48 hours. • If symptoms persist 48-72 hours after finishing antibiotic treatment or symptoms develop that are not symptoms of acute cystitis, patient should be advised to see a GP.
	Fosfomycin (2nd line) 3g orally, as a single dose (at night) (Supply 1 sachet)	Contraindications: • Previous serious adverse reaction or hypersensitivity to Fosfomycin • Severe renal impairment • Not recommended in cases of fructose intolerance, glucose-galactose malabsorption or sucrase-isomaltase insufficiency	
	Trimethoprim (3rd line) 300mg orally, daily (at night) for 3 nights (Supply 3 tablets)	Contraindications: • If the patient has been treated with trimethoprim in the previous 3 months, or had a trimethoprim-resistant E. coli isolate during this time, use an alternative antibiotic for empirical therapy. • Previous serious adverse reaction to trimethoprim-containing medicines • Megaloblastic anaemia due to folate deficiency • Other causes of folate deficiency • Other severe blood disorders • Porphyria • Hyperkalaemia • Treatment with methotrexate • Treatment with phenytoin • Treatment with lamivudine	

Patients on certain antibiotics (e.g., **nitrofurantoin and fosfomycin**) should **avoid the use of alkalinising agents** as they may significantly reduce the efficacy of their antibiotics.